

Table 1: Per-object contact fidelity and dynamic quality on CORE4D-G1 (all four methods share the same 52 sequences / 11 objects). $\uparrow$: higher is better, $\downarrow$: lower is better. **Bold** = best, underline = second best. Ours = SPIDER-CEM. Contact metrics are penetration-aware (a loose "any-contact" count that rewards interpenetration is intentionally omitted). Dynamic-quality metrics for Ours, SBTO and OmniRetarget are computed on a MuJoCo physics rollout (native for the two optimizers; position-servo replay for the kinematic OmniRetarget, E197 protocol) via the same code, hence directly comparable; GMR's are from its own kinematic pipeline at a different scale and are indicative only (read jointly with contact).

Object	n	Phys. contact@3mm $\uparrow$				Phys. pen@3mm $\downarrow$				Geom. pen@2mm $\downarrow$				Ankle jerk p95 (m/s ³) $\downarrow$				Obj speed max (m/s) $\downarrow$				Foot slip max (m) $\downarrow$			
		Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR
box001	6	0.612	<u>0.208</u>	0.015	0.000	<u>0.168</u>	0.366	0.607	0.009	<u>0.038</u>	0.302	0.507	0.009	<u>540.1</u>	840.0	1315.3	136.8	1.14	1.45	1.95	<u>1.29</u>	0.83	1.22	1.23	<u>0.85</u>
box004	3	0.486	0.005	<u>0.064</u>	0.000	<u>0.123</u>	0.448	0.571	0.000	<u>0.054</u>	0.448	0.541	0.000	<u>805.2</u>	1330.9	1555.5	184.1	1.36	<u>2.38</u>	3.51	2.40	1.10	0.74	1.13	<u>0.77</u>
box021	11	0.552	0.145	<u>0.149</u>	0.000	<u>0.180</u>	0.454	0.579	0.092	0.062	0.426	0.225	<u>0.092</u>	<u>707.7</u>	1223.1	1638.7	176.6	1.32	<u>1.98</u>	3.54	2.31	<u>1.10</u>	1.38	1.38	0.79
box023	7	0.583	0.194	<u>0.281</u>	0.000	<u>0.110</u>	0.327	0.337	0.013	<u>0.076</u>	0.327	0.272	0.013	<u>739.9</u>	1039.1	1743.0	212.6	1.52	2.18	3.05	<u>2.04</u>	<u>0.79</u>	1.07	1.25	0.57
box024	4	0.510	<u>0.289</u>	0.000	0.006	<u>0.264</u>	0.480	0.751	0.216	0.037	<u>0.194</u>	0.526	0.222	<u>613.8</u>	1132.4	1611.8	152.0	1.40	1.99	2.28	<u>1.48</u>	<u>1.04</u>	1.53	1.53	0.93
bucket003	3	0.564	<u>0.259</u>	0.009	0.002	<u>0.250</u>	0.452	0.544	0.019	<u>0.038</u>	0.433	0.378	0.019	<u>665.8</u>	1092.0	1547.2	178.9	1.36	1.73	2.12	<u>1.41</u>	0.65	<u>0.86</u>	1.52	1.51
bucket007	4	0.492	<u>0.156</u>	0.062	0.000	<u>0.202</u>	0.552	0.522	0.048	<u>0.055</u>	0.542	0.514	0.048	<u>691.5</u>	884.3	1324.3	167.6	1.17	1.64	2.32	<u>1.55</u>	0.60	<u>1.02</u>	1.14	1.27
chair006	4	0.711	<u>0.145</u>	0.038	0.000	<u>0.173</u>	0.633	0.472	0.000	0.045	0.536	0.367	<u>0.068</u>	<u>482.1</u>	851.7	1437.1	135.5	1.08	1.45	1.85	<u>1.22</u>	0.92	<u>1.14</u>	1.16	1.21
desk007	3	0.735	<u>0.263</u>	0.014	0.000	<u>0.154</u>	0.596	0.842	0.000	<u>0.062</u>	0.670	0.814	0.000	<u>527.0</u>	733.0	1181.6	106.9	1.06	1.34	1.56	<u>1.07</u>	0.52	1.19	<u>1.16</u>	1.39
desk021	4	0.619	0.100	<u>0.149</u>	0.004	<u>0.184</u>	0.654	0.346	0.006	0.026	0.553	0.180	<u>0.102</u>	<u>704.7</u>	1296.9	1716.8	159.1	1.28	<u>1.82</u>	4.28	2.87	0.89	<u>1.11</u>	1.79	1.43
desk023	3	0.685	<u>0.136</u>	0.101	0.005	<u>0.170</u>	0.553	0.421	0.028	<u>0.044</u>	0.489	0.285	0.040	<u>610.5</u>	1064.4	1943.9	246.8	1.31	<u>1.89</u>	3.59	2.36	<u>0.65</u>	0.93	1.36	0.64
Overall	52	0.588	<u>0.172</u>	0.101	0.001	<u>0.176</u>	0.479	0.536	0.046	0.052	0.427	0.381	<u>0.059</u>	<u>653.8</u>	1059.4	1560.8	170.4	1.29	<u>1.83</u>	2.83	1.88	0.87	1.16	1.33	<u>0.96</u>